$$Lu = u_t - [a_{ij}(x,t)u_{x_i x_j} + b_i(x,t)u_{x_i} + c(x,t)u]$$

WMP: L degenerately parabolic with $c \leq 0$ in a bdd domain $D \subseteq \mathbb{R}^{n+1}$

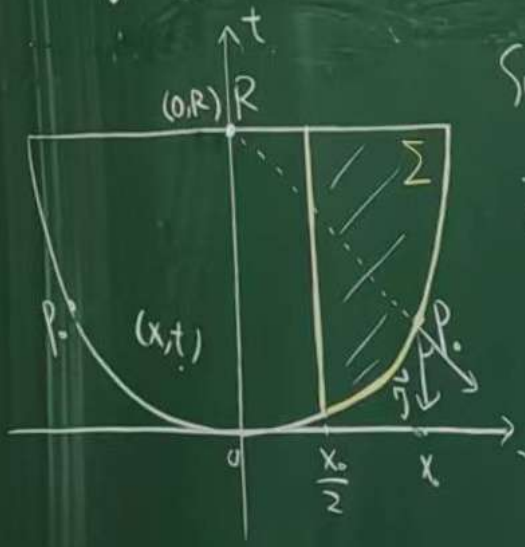
$u \in C^{2,1}(D \setminus \Gamma)$ s.t. $Lu \leq 0$ in $D \setminus \Gamma$

$\Rightarrow$ · $\max_{\bar{D}} u \leq \max_{\Gamma} u^+$

· $\max_{\bar{D}} u = \max_{\Gamma} u$ provided $\max_{\bar{D}} u > 0$ or $c \equiv 0$

2.3 Hopf boundary point Lemmas.

Baby Hopf B.P.L: $B_R^- = \{(x,t) \in \mathbb{R}^n \times \mathbb{R} \mid |x|^2 + (t-R)^2 < R^2,\ 0 < t \le R\}$ $(\in \mathcal{D}(\Gamma))$



Suppose L is strictly parabolic and a_{ij}, b_i, c bdd in B_R^-. Let $u \in C^{2,1}(B_R^-) \cap C^0(\overline{B_R^-})$ with $Lu \le 0$ in B_R^-.

Assume $\exists\ P_0 = (x_0, t_0) \in \Gamma$ s.t. $x_0 \neq 0$;

$u(x_0,t_0) > u(x,t)\ \forall\ (x,t) \in \overline{B_R^-} \setminus \{P_0\}$.

Then $\forall$ outward pointing vector $\vec{\nu}$ at P_0, we have

$\boxed{\frac{\partial u}{\partial \vec{\nu}}(P_0) > 0}$ (if it exists),

provided

(i) $(\equiv 0$ or (ii) $(\le 0$ in B_R^- & $u(P_0) \ge 0$, or (iii) $u(P_0) = 0$.

WMP $\Rightarrow \max_{\overline{\Sigma}} w \le \max_{\Gamma_\Sigma} w^+ = 0 \Rightarrow w \le 0$ in Σ

Note $w(P_0) = 0$, thus $\frac{\partial w}{\partial \vec{\gamma}}(P_0) \gtrless 0 \Rightarrow \frac{\partial u}{\partial \vec{\gamma}}(P_0) = \underbrace{\frac{\partial w}{\partial \vec{\gamma}}(P_0)}_{\gtrless 0} - \underbrace{\varepsilon \frac{\partial r}{\partial \vec{\gamma}}(P_0)}_{>0} > 0$.

If (iii) holds, then $u \le 0$ in B_R^-.

$$0 \gtrless Lu = \underbrace{u_t - \left[a_{ij} u_{x_i x_j} + b_i u_{x_i} + c^- u\right]}_{:= \hat{L}u} + \underbrace{c^+ u}_{\le 0}$$

$\Rightarrow \hat{L}u \le 0$ in $B_R^- \overset{\text{(ii)}}{\Longrightarrow} \frac{\partial u}{\partial \vec{\gamma}}(P_0) > 0$.

Pf: Assume (i) and (ii) hold first.

We will construct $v \in C^\infty(\mathbb{R}^{n+1})$ s.t.

(a) $Lv < 0$ in $\Sigma = \left\{ (x,t) \in B_R^- \;\middle|\; |x| > \frac{|x_0|}{2} \right\}$.

(b) $v|_\Gamma = 0$. (c) $\frac{\partial v}{\partial \vec{\gamma}}(P_0) < 0$.

Then define $w(x,t) = u(x,t) - u(P_0) + \varepsilon v(x,t)$, $\varepsilon > 0$ small.

Now, apply WMP (≤ 0) on Σ to w:

$$Lw = \underbrace{Lu}_{\leq 0} + \varepsilon \underbrace{Lv}_{<0} + \underbrace{c(x,t) \cdot u(P_0)}_{\leq 0 \text{ (under (i) \& (ii))}} < 0 \quad \text{in } \Sigma.$$

$w|_{\Gamma \cap \Gamma_\Sigma} = u - u(P_0) \leq 0$, $w|_{\Gamma_\Sigma \cap \{|x| = \frac{|x_0|}{2}\}} \leq 0$ ($\because u(x,t) - u(P_0) \leq -\delta < 0$

Construction of V: $V(x,t) = e^{-\alpha(|x|^2+(t-R)^2)} - e^{-\alpha R^2}$ ($\alpha>0$ to be chosen)

$= E(x,t) - e^{-\alpha R^2}$

- (b) ✓
- (c): ✓

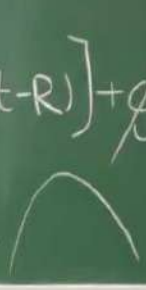
$$V_{x_i} = E(x,t)(-2\alpha x_i), \quad V_t = E(x,t)\left[-2\alpha(t-R)\right]$$

$$\frac{\partial V}{\partial \vec{\jmath}}(P_0) = \nabla_{(x,t)} V(P_0)\cdot\vec{\jmath} = E(x,t)\cdot\left[-2\alpha(x_0, t_0-R)\right]\cdot\vec{\jmath} < 0.$$

- (a): $V_{x_i x_j} = E(x,t)\,4\alpha^2 x_i x_j - E(x,t)(2\alpha)\delta_{ij}$

$$LV = E(x,t)\cdot\left[-4\alpha^2 a_{ij}x_i x_j + 2\alpha\textstyle\sum a_{ij} + 2\alpha\sum b_i x_i - c(x,t) - 2\alpha(t-R)\right] + \underbrace{c(x,t)e^{-\alpha R^2}}_{\le 0}$$

$$\le E(x,t)\cdot\left[-\lambda_0\|x_0\|^2\alpha^2 + (2nM + 2\sqrt{n}MR + 2R)\alpha + M\right].$$

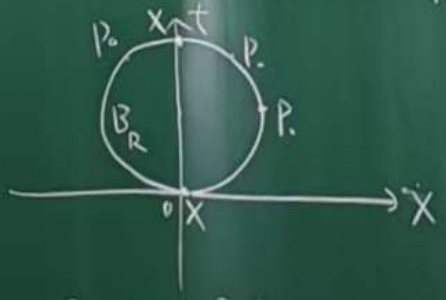


where λ_0 satisfies $\left(a_{ij}(x,t)\right)_{n\times n} \ge \lambda_0 I_{n\times n}$ $\forall\ (x,t)\in B_R^-$.

M satisfies $\|a_{ij}, b_i, c\|_{L^\infty(B_R^-)} \le M$

$LV<0$ in B_R^- provided $\alpha>0$ is sufficiently large. □

RK: 1. This Lemma is true if $\bar{B_R}$ is replaced by B_R, provided P_0 is neither the South pole nor the North pole. D.





2. True if $\bar{B_R}$ & B_R are shifted but not rotated.

Hopf B.P.L (grown-up version):

Suppose that L is strictly parabolic with a_{ij}, b_i, c being bdd in $D \subseteq \mathbb{R}^{n+1}$. Assume $u \in C^{2,1}(\bar{D} \setminus \Gamma)$ s.t. $Lu \leq 0$ in $\bar{D} \setminus \Gamma$.

If $P_0 \in \Gamma$ is a strict local max, u is continuous at P_0, where interior sphere condition is satisfies, and P_0 is neither the South pole nor the north pole, then

$$\frac{\partial u}{\partial \vec{n}}(P_0) > 0 \quad \text{(if it exists)} \quad \forall \text{ outward pointing } \vec{n} \text{ at } P_0;$$

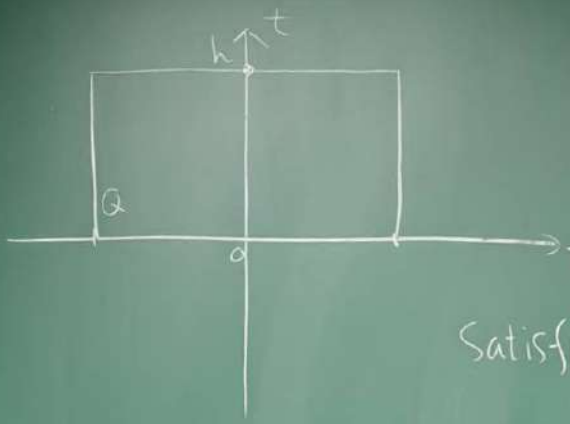
provided

(i) $c \equiv 0$, (ii) $c \leq 0$, $u(P_0) \geqslant 0$, (iii) $u(P_0) = 0$.



$u(P_0) > u(x,t) \quad \forall \ (x,t) \in \mathcal{U} \cap \bar{D}$

Lemma Let $Q(R,h)=\{(x,t)\in\mathbb{R}^n\times\mathbb{R} \mid |x|<R,\ 0<t\le h\}$



Suppose L is strictly parabolic with a_{ij}, b_i, c being bdd in $Q(R,h)$

Then there exists no $u\in C^{2,1}(Q)\cap C^0(\bar{Q})$

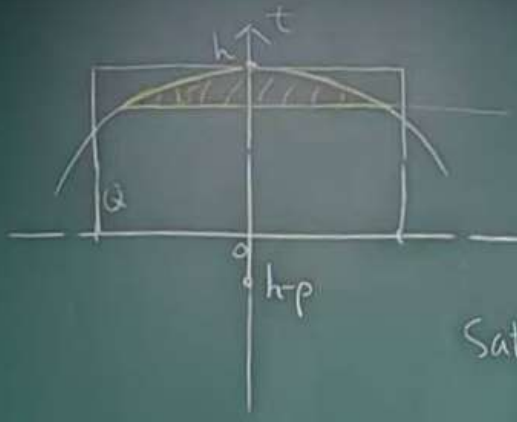
satisfies ① $Lu\le 0$ in $Q(R,h)$;

② $u(0,h)>u(x,t)$ $\forall$ $|x|<R$, $0<t<h$;

③ at least one of the following holds

(i) $c\equiv 0$; (ii) $c\le 0$ & $u(0,h)\ge 0$, (iii) $u(0,h)=0$.

Lemma Let $Q(R,h)=\{(x,t)\in\mathbb{R}^n\times\mathbb{R} \mid |x|<R,\ 0<t\le h\}$



Suppose L is strictly parabolic with a_{ij}, b_i, c being bdd in $Q(R,h)$

Then there exists no $u\in C^{2,1}(Q)\cap C^0(\bar{Q})$

satisfies ① $Lu\le 0$ in $Q(R,h)$;

② $u(0,h) > u(x,t)$ $\forall$ $|x|<R$, $0<t<h$;

③ at least one of the following holds

(i) $c\equiv 0$, (ii) $c<0$ & $u(0,h)\ge 0$, (iii) $u(0,h)=0$.

Pf: Claim: Assume such u exists. Then $\frac{\partial u}{\partial t}(0,h) > 0$.

However, $Lu(0,h) = \underbrace{\frac{\partial u}{\partial t}(0,h)}_{>0 \text{ (by claim)}} - [\underbrace{a_{ij}(0,h) u_{x_i x_j}(0,h)}_{\leq 0} + \underbrace{b_i(0,h) u_{x_i}(0,h)}_{=0} + \underbrace{(cu)(0,h)}_{\leq 0 \text{ (by (i)-(iii))}}]$

$> 0 \quad \not\leq 0$ ①

Pf of claim: Take a $\rho > 0$ and $\delta > 0$ s.t.

$$N := \overline{B_\rho(0,h-\rho) \cap \{t > h-\delta\}} \subset Q(R,h)$$

Will construct $V \in C^\infty(\mathbb{R}^{n+1})$ s.t.

(a) $Lv<0$ in N (b) $v|_{\partial B_\rho(0,h-\rho)}=0$ (c) $\frac{\partial v}{\partial t}(0,h)<0$

Then let $w(x,t)=u(x,t)-u(0,h)+\varepsilon v(x,t)$

· $Lw = \underbrace{Lu}_{\le 0} + \underbrace{c(x,t)u(0,h)}_{\le 0} + \varepsilon \underbrace{Lv}_{<0} < 0$ in N.

((i) or (ii))

· $w \le 0$ on ∂N

WMP ⟹ (with $c \le 0$) If (i) or (ii) holds, $\max_{\bar{N}} w \le \max_{\partial N} w^+ = 0$.

$\Rightarrow w \le 0$ in $\bar{N}$, $w(0,h)=0$ $\Rightarrow \frac{\partial w}{\partial t}(0,h) \ge 0 \Rightarrow \frac{\partial u}{\partial t}(0,h) > 0$.

If (iii) holds, then $u \le 0$ in $Q(R,h)$

$$0 \ge Lu \ge \underbrace{u_t - [a_{ij}u_{x_i x_j} + b_i u_{x_i} + c^- u}_{\hat{L}u} + \underbrace{c^+ u}_{\le 0}] \Rightarrow \hat{L}u \le 0$$

$$\overset{\text{(ii)}}{\Rightarrow} \frac{\partial u}{\partial t}(0,h) > 0$$

Construction of V under (i) or (ii):

$$V(x,t) = \rho^2 - |x|^2 - (t-(h-\rho))^2 = -|x|^2 - (t-h)^2 - 2(t-h)\rho \le -2(t-h)\rho \le 2\delta\rho \quad \text{in } N$$

(b) ✓

(c) $\frac{\partial V}{\partial t}(0,h) = -2\rho < 0$ ✓

(a) $LV = -2(t-h+\rho) + 2\sum a_{ij}\delta_{ij} + 2\sum b_i x_i - c(x,t)V$

$$\leq -2(h-\delta-h+\rho) + 2nM + 2\sqrt{n}\,MR + 2\delta\rho M$$

$$= 2\rho(-1+\delta M) + 2\delta + 2M(n+\sqrt{n}R)$$

< 0 if ρ is big, δ is small.